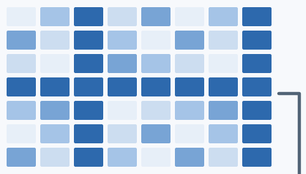
FastVCL Sparse-to-tube video copy localization

PROPOSED METHOD · SCHEMATIC

1 Fixed features

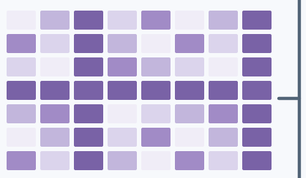
Query A / reference B

Query A



F_A: M × 256

Reference B

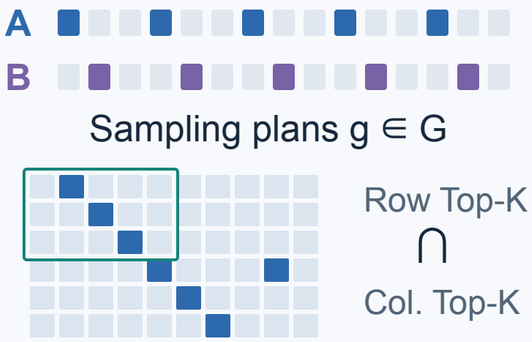


F_B: N × 256

L2 normalization
+ validity masks

2 Sparse matching

Sample → compare → retain

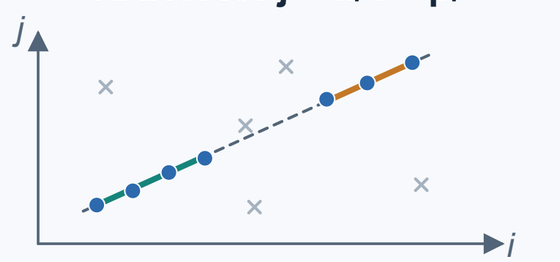


Deduplicate coordinates; cap

Anchor set {(i, j, S_{ij})}

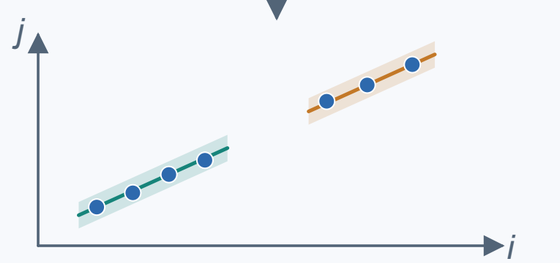
3 Temporal proposals

Fit → validate → split



Distinct-time support on A and B

Split long gaps → refit / validate



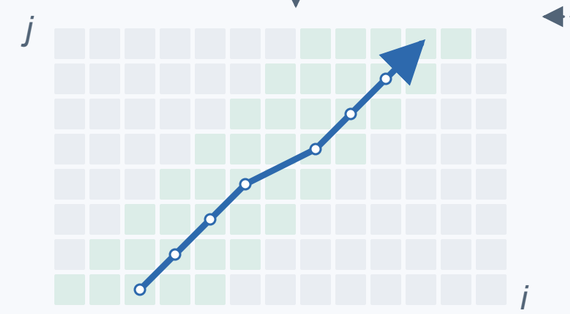
Extend A: e_A B band: ±w_r

Finite T_r: $\ell_r \le i < u_r$

4 Local refinement

Align → recover → candidates

Cosine scores only inside T_r



Local DP with a zero restart

$H_{ij} = \max\{0, r_{ij}, \text{continuation}\}$

Traceback → trim → score

↻ Block matched cells; repeat



Score q · min(IoU_A, IoU_B) dedup.

A0

A1

C0

ONE SHARED MODEL · TWO CALL SITES

1 Anchor decision

A0 → budgeted hard-sample selection
NOT_COPY + margin → drop
COPY / uncertain / fail → keep
A1 = A0 \ D(a), then refit

seltive model

one shared external model
COPY / NOT_COPY / UNCERTAIN
MERGE / KEEP_SEPARATE
abstain · failure keeps evidence
one budget K · cache · one revision

2 Fusion decision

C0 → legal edges (G, gaps)
same copy? + Conn
M = G ∧ Conn ∧ margin
merge / keep · one round
→ Y final intervals

A/B original frames + verified index → frame map

Full-resolution normalized descriptors + validity masks

● anchor — track ▭ tube → local path

Core + two budgeted visual decisions · feature-index coordinates